

1. Group Information

❖ **Group Name:** Tech Titans

❖ **Group Members:**

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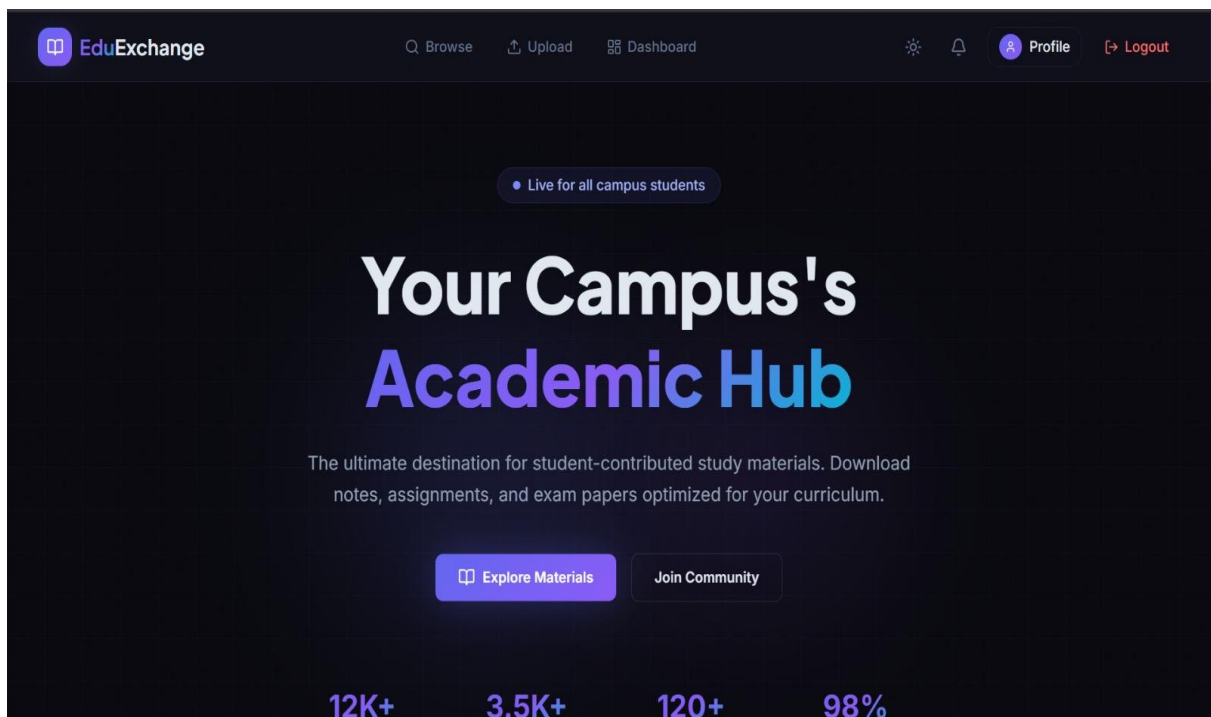
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2. Title of the Project

Digital Platform for Seamless Campus Material Exchange



3. Project Pitch

Edu Exchange is a cloud-based web platform designed to simplify the sharing and access of academic resources among university students. Built using the MERN stack (MongoDB, Express.js, React, and Node.js) with AWS S3 and Firebase integration, it provides a centralized and secure space for uploading, storing, and downloading study materials. The platform eliminates the limitations of traditional note-sharing methods such as scattered messaging groups and limited physical copies. With features like intelligent categorization, advanced search functionality, and JWT-based authentication, Edu Exchange enables efficient collaboration and ensures equal access to quality academic resources for all students.

4. Problem Statement

- In today's higher education environment, students often face significant challenges in accessing reliable, organized, and updated academic resources. The problem becomes more critical during mid-semester and end-semester examinations when the demand for notes, previous-year papers, assignments, and study materials increase rapidly. Currently, most resource sharing depends on informal methods such as WhatsApp groups, Telegram channels, personal drives, and physical photocopies. These methods are inefficient and create several obstacles for students.
- One major issue is the loss of important academic data. Study materials shared through messaging applications often get buried in long chat histories, making them difficult to retrieve when needed. Additionally, cloud storage links may expire or become inaccessible over time, resulting in the permanent loss of valuable resources. New students joining a class or group are often unable to access previously shared documents.
- Another challenge is unequal access to academic materials. Students who have strong peer networks can easily obtain notes and study resources, while others may struggle to find the same information.

Physical notes and limited library resources can also be monopolized by a small number of students, creating an unfair learning environment.

- Furthermore, there is a lack of a centralized and structured repository for academic content. Finding subject-specific notes, previous-year materials, or important documents from earlier batches is often a time-consuming and frustrating process. The absence of proper categorization and search functionality further reduces efficiency.
- Edu Exchange addresses these challenges by providing a secure, centralized, and cloud-based platform dedicated to academic resource sharing. The system enables students to upload, organize, search, and download study materials efficiently. By creating a permanent digital repository, Edu Exchange promotes equal access to educational resources, enhances collaboration among students, and reduces the time spent searching for academic content, allowing learners to focus more effectively on their studies.

5. Project Aim and Objectives

Aim of the Project:

The primary aim of the Edu Exchange project is to develop a centralized cloud-based platform that enables students to securely upload, organize, search, and download academic resources. The system provides easy access to study materials such as notes, assignments, and previous-year question papers. It aims to improve collaboration among students and create a reliable digital repository for educational content. The platform also helps reduce the time spent searching for resources and promotes equal access to learning materials for all students.

Objectives of the Project:

1. To develop a centralized platform for sharing and accessing academic resources.
2. To provide secure user authentication and authorization using Firebase Authentication and JWT.

3. To enable students to upload, download, and manage study materials efficiently.
4. To create a responsive and user-friendly interface using React and Tailwind CSS.
5. To implement a robust backend using Node.js and Express.js for efficient data processing.
6. To integrate cloud storage services for secure and scalable file storage.
7. To organize resources based on course, semester, subject, and resource type.
8. To provide advanced search and filtering features for quick resource discovery.
9. To encourage collaboration and knowledge sharing among students through a common platform.
10. To ensure system scalability, reliability, and high performance during peak usage periods.

6. Project Methodology

i. Project Identification & Research

The initial phase involved deep requirement engineering. We analyzed the explicit pain students face when gathering study materials. Surveys and interviews were conducted within the BCA department to determine the most demanded file formats, the most stressful academic periods, and the failures of existing informal systems. This empirical data formed the foundation of our centralized digital concept.

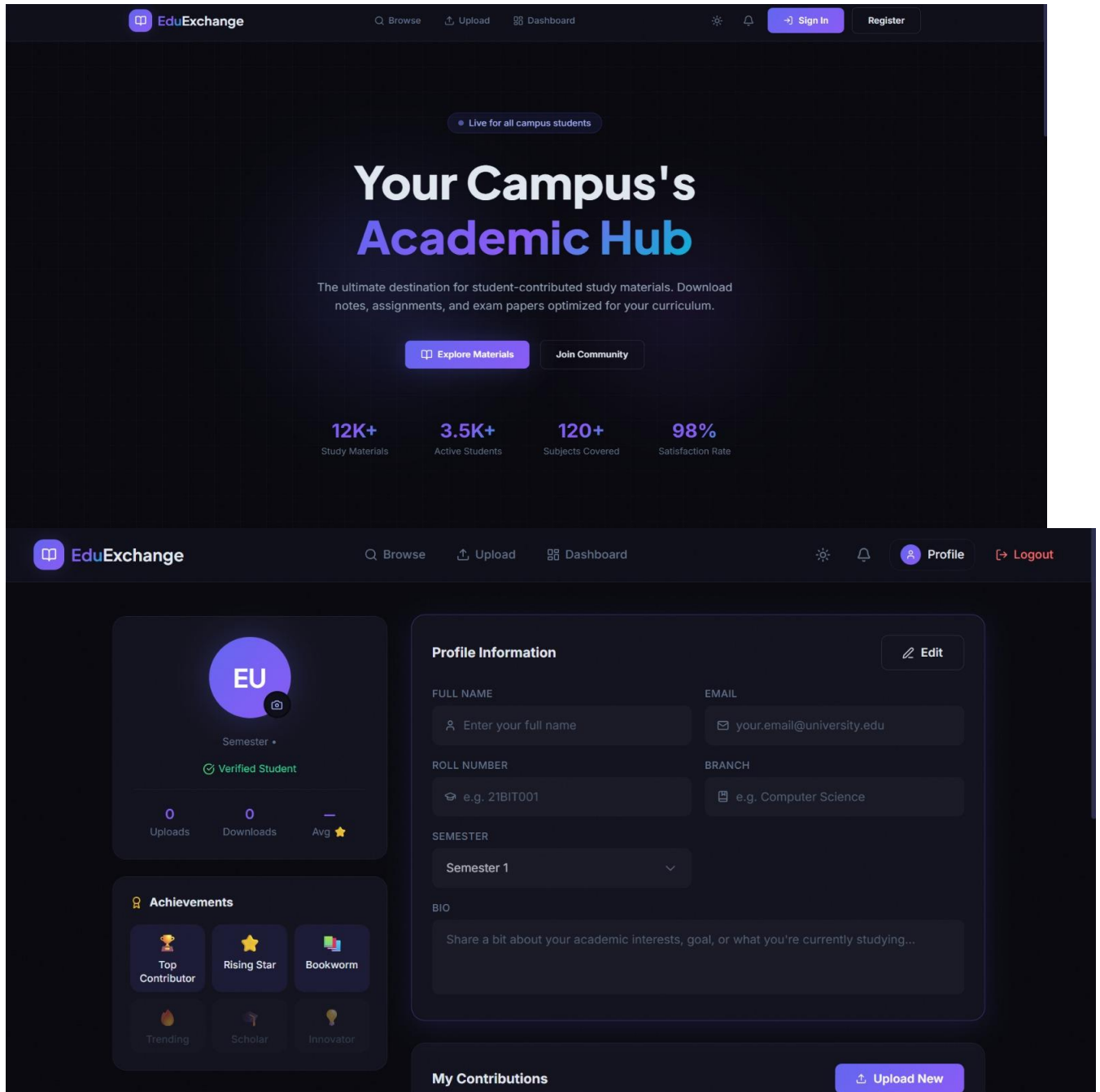
ii. Design & Planning:

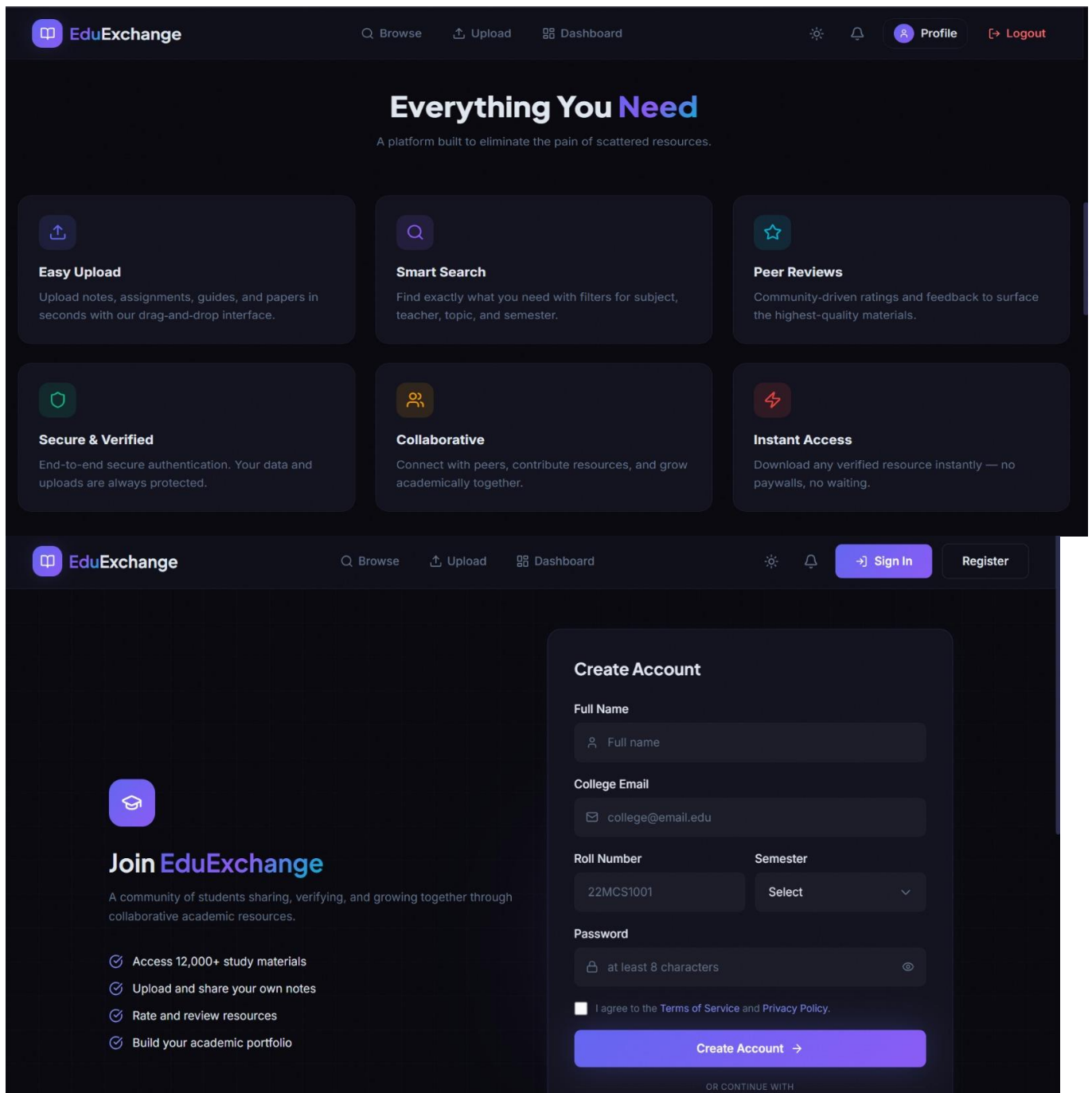
Before writing any code, comprehensive UI wireframes and user flow diagrams were drafted using modern design principles. The complex database schema was meticulously mapped out for MongoDB Atlas, ensuring relational integrity within a NoSQL environment. We completely planned the RESTful API endpoints and established the secure routing protocols leveraging JWT middleware

iii. H/W and S/W Component Selection

A conscious decision was made to omit traditional, physical on-premises hardware entirely, leaning fully into modern cloud architecture. We selected the MERN stack (MongoDB, Express, React, Node.js) for its unmatched synergy. Tailwind CSS was chosen for rapid frontend styling. AWS S3 and Firebase were selected as the undisputed leaders in secure, scalable cloud storage.

iv. Prototype Development





v. Testing and Optimization

